

Supplementary information

Genome editing in almond: A CRISPR-based approach through hairy root transformation

Supplementary Table S1. List of oligonucleotides

Supplementary Table S2. List of all potential off-target sites predicted by the Cas-OFFinder tool in the almond genome (*P. dulcis* cv. Texas)

Supplementary Table S1. List of oligonucleotides.

Guides

(Esp31 restriction sites in red lower-case, cut site in red upper-case, guide sequence in green)

ERF74_guide1_fw	GTGcgtctcAATTGGCTTTCCAGCAGAGAAAGCAGGTTTGgagacgCAC
ERF74_guide1_rev	GTGcgtctcCAAACCTGCTTTCTCTGCTGGAAAGCCAATTgagacgCAC
ERF74_guide2_fw	GTGcgtctcAATTGGGAATTCGCCAGCGCCCATGGTTTGgagacgCAC
ERF74_guide2_rev	GTGcgtctcCAAACCATGGGCGCTGGCGAATTCCCAATTgagacgCAC
GAI_guide1_fw	GTGcgtctcAATTGGTACCGTTCACTATAACCCGTGTTTGgagacgCAC
GAI_guide1_rev	GTGcgtctcCAAACACGGGTTATAGTGAACGGTACCAATTgagacgCAC
GAI_guide2_fw	GTGcgtctcAATTGGCAACATATCGGAGAATGAGGTTTGgagacgCAC
GAI_guide2_rev	GTGcgtctcCAAACCTCATTTCTCCGATATGTTGCAATTgagacgCAC

Target loci amplification

ERF74_locus1_fw	TGTGGAGGTGCTATAATATCCGATTTTC
ERF74_locus1_rev	TCTCTAACATCAATCCACCCCTTTAAACC
ERF74_locus2_fw	CTCTACGACTGTGAAATCAGTGGAGTTC
ERF74_locus2_rev	CACTGAGTTGCCACCATCCTGAG
GAI_locus1_fw	TATCCCTATCCAGACCCGTCATC
GAI_locus1_rev	GAAGACGAGGTGAGAGGCAGAAGAC
GAI_locus2_fw	AAACCCACACGATTCTTCGTCC
GAI_locus2_rev	TCGACTCAGTGAGCCATCACCG

RT-qPCR

Q ERF74_fw	GGAAAGCCCTCTTCTGCTCGTA
Q ERF74_rev	CCTTGGGTCTCGGATCTCTGC
Q ACT_fw	TTGGAATGGAGGCTGCTGGTATT
Q ACT_rev	GCCTCCAATCCAGACGCTGT

Off-target analysis

ERF74_g1_Pd01_-16219319_fw	CCTTGGCTCATATACCTCATCACAT
ERF74_g1_Pd01_-16219319_rev	GTAAATCTCCCTACATTCTCTATGC
ERF74_g1_Pd03_-9068856_fw	CCAAGAACCTGCAAGAAGACATTG
ERF74_g1_Pd03_-9068856_rev	TCCTGAACTCATCTTTGGTG
ERF74_g1_Pd04_-9549693_fw	AAACTAAAGCTCACGCTCGGG
ERF74_g1_Pd04_-9549693_rev	GGATAACAATACCATCCTCGGC
ERF74_g2_Pd04_-22567141_fw	TTACAGTGGAGATCATTTCCCAAGC
ERF74_g2_Pd04_-22567141_rev	GAAGATAACGCCACACGCAGAC
ERF74_g2_Pd06_+3018280_fw	GACTCCTCCAGCGACGAAGAAG
ERF74_g2_Pd06_+3018280_rev	CGGCGTTAGGAAAGTTGGTGAC
ERF74_g2_Pd06_+3042881_fw	GACTCCTCCAGCGACGAAGAAG
ERF74_g2_Pd06_+3042881_rev	CGGCGTTAGGAAAGTTGGTGAC
GAI_g1_Pd07_+4850919_fw	CGAAGTAGCCTACCCTCTCAACTG
GAI_g1_Pd07_+4850919_rev	TCAGAAATCCTGAATCCGCCAT
GAI_g2_Pd01_+8709365_fw	GAGGCTCAAACGGTATCCAAA
GAI_g2_Pd01_+8709365_rev	TGACGAGTAAGGTATGGATTCTAC
GAI_g2_Pd02_-14236738_fw	GAGTAGAATCAGAGAAGGTGCTTG
GAI_g2_Pd02_-14236738_rev	CGCAATGATTCCATCTCTCTTT
GAI_g2_Pd08_-6198750_fw	TGCTTCTTCTCAATGCTATATGTGC
GAI_g2_Pd08_-6198750_rev	CTAACATCCGATAAATGCTTGGTG

Supplementary Table 2. List of all potential off-target sites predicted by the Cas-OFFinder tool in the almond genome (*P. dulcis* cv. Texas). The PAM motif is highlighted in bold, and the mismatched bases to the original target sites are shown in red lowercase letters. Mutation rate represents the number of hairy roots with mutations divided by the total number of tested hairy roots. MMs, number of mismatches. ^a, homologous regions (250 bp up- and down-stream to the target site). ^b, amplified off-target site in the studied cultivar Vairo that differed from the *in silico* predicted off-target in cultivar Texas. Consequently, the corresponding genomic regions were sequenced in wild-type roots of the studied cultivar and compared to those from hairy roots. n.a., not analyzed.

gRNA	Putative off-target locus	Sequence	MMs	Mutation rate
ERFguide1	Pd01: -16219319	aTTTCCAGCAGAA AAAA CAG TGG	3	0/8
	Pd03: -9068856	gTTTCCAGgAGAGAA tGCAGTGG	3	0/8
	Pd04: -9549693	agTTCCAGCAGAGAAgGCAG GGG	3	0/8
	Pd06: -23078226	aTTT ga AtCAGAGAAAGCAG AGG	4	n.a.
	Pd08: +3965791	agTTCCAGC tGAGAAgGCAGGGG	4	n.a.
	Pd03: -9062785	gTTTCCAGgAGAGg tGCAGTGG	4	n.a.
	Pd03: -6823832	gTTTCCAGaAGAGAAAGg aAGG	4	n.a.
	Pd07: -17360615	tTTTCCAGCAGAc AAgGCAaAGG	4	n.a.
	Pd03: +12342957	Ca TCC tGCAGAGAAAGC tGGG	4	n.a.
	Pd04: -4643756	CT ctG CA aCAGtGAAAGCAGCGG	4	n.a.
	Pd04: -22874809	CTg ttt AGCAGAA AAAGCAGGGG	4	n.a.
	Pd03: +12628686	CTg aCC AGCA aAGAAAGaAGAGG	4	n.a.
	Pd05: +6915268	CTTT t Ca tCAaAGAAAtCAGTGG	4	n.a.
	Pd05: -6066496	CTTT aCAaa aCAGAAAGCAG AGG	4	n.a.
	Pd05: -6071075	CTTT aCAaa aCAGAAAGCAG AGG	4	n.a.
	Pd02: +12590413	CTTTg CaCC AGA agAAGCAGAGG	4	n.a.
	pdulcis26_s0686: +62241	CTTTg CaCC AGA agAAGCAGAGG	4	n.a.
	Pd07: +759994	CTTTg CAGC AGAG AttacAGGGG	4	n.a.
	Pd08: -17626304	CTTT CaAGC AGAG tAGgcGTGG	4	n.a.
	Pd04: +7685049	CTTg gCAGCtGAGAAAGCtGAGG	4	n.a.
	Pd06: -28376323	CTTg CaAGCAaAGAAAaCAGAGG	4	n.a.
	Pd05: +1381635	CTT ccc AtCAGAGAg AGtAGTGG	4	n.a.
	Pd08: +11186703	CTT ccc AGCA aAgAAGCAGTGG	4	n.a.
	Pd03: +18792526	CTT ccc AGCAG tGAAAcCtGGG	4	n.a.
	Pd03: -4591175	CT aT CaAGCAGAGAA aCtGGG	4	n.a.
	Pd08: +18797158	C TT CCAGaAG AtAAAGCtGTGG	4	n.a.
ERFguide2	Pd04: -22567141	GGAg tCCG CaAGCG gCCATGGG	4	0/8
	Pd06: +3018280	GGAg tCCG CCgGCG aCCATGGG	4	0/8 ^a
	Pd06: +3042881	GGAg tCCG CCgGCG aCCATGGG	4	0/8 ^a
GAIguide1	Pd07: +4850919	TACa GTTCg CTATAc ACCGTTGG	4	0/2 ^b
GAIguide2	Pd01: +8709365	aaA ActgATCGGAGAATGAG TGG	4	0/2 ^b
	Pd02: -14236738	GCA AA TAT aaGA AAATGAG AGG	4	0/2 ^b
	Pd08: -6198750	GC atg ATATCG aAGA ATGA aAGG	4	0/2